

Experience

JUMP TRADING starting Jun 2026
Software Engineering internship.

NTT DATA (via UCL IXN) Oct 2025 – Mar 2026
Carbon-aware AI workload scheduling. Optimal resource allocation webservice implemented in C++23 with policy-based parallelism, x86 SIMD, and coroutines. [Repo](#)

UCL Sep 2025 – Mar 2026
Teaching Assistant and my course's Programming Tutor.

Scholastic

UCL COMPETITIVE PROGRAMMING SOCIETY Nov 2024 – Present
Founding & committee member and speaker. We host topical lectures, contests, and training sessions with 130 members.

ACADEMIC RESEARCH May 2023 – Oct 2023
Co-authored a [16-page report](#) on water usage in the UAE. Presented at the [Roots & Shoots Expo Awards Ceremony](#).

JESS AI CLUB Jan 2022 – May 2022
Founder and host of the first purely student-ran club. Its popularity allowed me to introduce NLP theory to 20+ peers.

Competitions

COMPETITIVE PROGRAMMING Jan 2025 – Present
[11th](#)/310 (1st in London) at the 2025 [UKIEPC](#)
[53rd](#)/141 at the 2025 [ICPC NWERC](#)
2nd at the 2025 [Recraft Mini-Cup](#)
Codeforces: 16th in UK (Mar 2026). Leetcode: top 7.61%.

MATHEMATICAL MODELING Nov 2022; Feb 2023
Awarded *Meritorious* in [HiMCM](#) and *successful in triage* in [M3C](#). Led and trained my own team. Performed time-series analysis, non-linear and periodic regression, and dynamics modeling using SciPy and Mathematica.

Projects

[gws2](#) (akioweh.com) launched Oct 2024
A compact server on continuous deployment prototyped with modern technologies like 0-RTT h3 and reverse proxying.

Team-based Software Engineering Oct 2025 – Dec 2025
An academic management system using Spring Boot & JPA with React developed concurrently as a team of three.

[OiqBot](#) launched Oct 2021
Real-time network app with a cooperative concurrency model and modular, hot-loadable plugins. Engineered a non-blocking (de)initialization procedure for quick restarts.

[sheepcalc](#) Jul 2025 – Jul 2025
Pure λ -calculus interpreter in Haskell using ADS and Parsec.

[AEQ](#) May 2024 – Mar 2025
Audio EQ manipulation tool. Supports parametric cascaded-biquad IIR and phase-controlled convolution FIR.

Education

UCL 2024 – 2027
BSc Computer Science Y1: 83.38%

JESS Dubai 2020 – 2024
GCSEs (11), IB Diploma

Specific Knowledge

ALGORITHMS & DATA-STRUCTURES
Usage in competition code & practical software; C++ template programming; Low-level performance optimization & benchmarking

SYSTEMS DESIGN
Cooperative concurrency;
Asynchronous business logic;
Microservices & monolithic topologies

LANGUAGES
Python (^{5yr}): ASGI, asyncio, typing
C++ (^{3yr}): **c++23**: ranges, coroutines, concepts, value semantics; CMake
Java (^{3yr}): Maven; JSP & JSTL, Spring
Haskell (^{1yr}); Lua (^{1yr})

DEVOPS
Infrastructure-as-code (Azure & OC);
CI/CD (GitHub)

DEV TOOLING
Git; Remote dev., "As-prod" envs via containers & package locking

TYPESETTING
Articles, booklets, and graphics in
Typst; $\text{\LaTeX}$

COLLABORATION
Misc. open-source contributions (see [GitHub profile](#))

Misc

LIVE AUDIO & VIDEO
Event audio-visual control & setup;
Live mixing and audio engineering

HOBBY PRODUCTS
[Split, Column-stagger 40% Keyboard Editor Configuration](#)

NATURAL LANGUAGES English, Mandarin